

MIN SEOHYEON

SOFTWARE ENGINEER

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Software engineer who builds systems in C++ and Unity — object/component architectures, state-machine and pattern-based gameplay systems, collision and rendering pipelines — then debugs and optimizes them under real performance constraints.

SKILLS

Programming	C++, C#, C, Python, JavaScript; gameplay/engine architecture, design patterns (Template Method, Composite), collision, debugging, memory/lifetime fixes
Systems	Custom C++ engines (raylib, OpenGL), Unity gameplay systems, boss/pattern frameworks, state and data-driven design, performance debugging
Workflow	Git branching and merge review, GitHub Projects/Issues, Notion, CMake, Visual Studio, WSL, profiling, technical specification

EXPERIENCE

TEACHING ASSISTANT - GAME DEVELOPMENT PROJECT I

DIGIPEN KOREA | SPRING 2025

- Supported ~30 students across DigiPen Korea's Game Development Project I cohort with C++ implementation, debugging, and technical problem-solving throughout the term.
- Diagnosed issues across student projects and delivered clear, actionable technical feedback to help teams identify problems and improve their implementations.

MANZO 🏆

Sep 2024 – Dec 2025

Graphics / Engine Programmer · Technical Artist · Production Lead · C++ / OpenGL / GLSL

- Implemented BPM timing windows, beat/bar counting, and audio-synchronized player movement and boss patterns.
- Built layer-sorted draw queues and a framebuffer post-processing pipeline for bloom, underwater distortion, god rays, and transitions.
- Diagnosed severe boss-fight frame drops to per-frame redundant collision checks and eliminated them, restoring stable performance.
- Eliminated dangling-pointer failures by moving scenario/dialogue ownership into engine-level systems. Largest repository contributor: 366 commits.

NEW MANZO 🏆

Aug 2025 – Sep 2026

Primary C# Programmer · Game Artist / Visual Director · Production Lead · Unity / C#

- Designed a Template Method-based boss pattern architecture that fixes shared logic across 21 concrete implementations.
- Built a Composite orchestration layer that chains sub-patterns into runtime combos, including combos nested inside combos.
- Decoupled projectile motion and spawn behavior into standalone interfaces independent of any concrete projectile.
- Primary C# contributor with 418 of 585 repository commits.

DOUBLE HIT 🏆

Mar – Jul 2024

Systems Programmer · Art / Audio · Production Lead · C++ / raylib

- Implemented a custom sprite-file parser that loads textures, animation frames, hotspots, and collision shapes from data.
- Replaced ~30 hardcoded platform coordinates with a PlatformManager parsing level geometry from a custom .plf format.
- Built the GameObjectManager driving per-frame update/draw and pairwise collision dispatch across all live objects.
- This implementation became the base Manzo later expanded into rendering, CCD, rhythm, and scenario systems.

BIRD STRIKE 🏆

Nov – Dec 2023

Gameplay Programmer · Game Designer · Artist · Production Lead

- Implemented audio-timeline beat detection and rhythm-synchronized crow spawning as the core gameplay loop.
- Built dynamic attack subdivision, player/crow movement, and atan2-based direction logic in a custom engine.
- Produced original art and audio as sole developer on a two-month project without a commercial engine.

TOO HOT! 🏆

Jul 2026

Technical Art · Creative / Technical Direction · Producer · Unity / ShaderLab

- Specified GameManager and per-stage ScriptableObject data flow, save-range safeguards, and chapter selection.
- Reviewed and tested teammate-authored gameplay implementations and clean-state debug controls.
- Balanced direct art/technical-art execution with a 130+ item P0-P3 backlog and two-programmer coordination.

STREET TYPER 🏆

Aug 2026

Art · Technical Art · Producer (Team Project) · Unity URP

- Specified, evaluated, and integrated a reusable UI shader workflow for rounded forms, gradients, and blur.
- Owned original 2D art, UI composition, particles, and hit VFX; gameplay code was teammate-authored.
- Published a playable build on itch.io and prepared the game for a Steam release.

EDUCATION

B.S. Computer Science in Real-Time Interactive Simulation | Expected Graduation: May 2028
Keimyung University | DigiPen Institute of Technology | GPA 3.958/4.0